Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

# SDC Hackathon

## Phase 2: Final Project & Prototype Report

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline:** 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Final.pdf” together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

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# 1. TEAM & PROJECT OVERVIEW

## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | T010NG |
| **Project / Product Name** | Kotak |
| **Final Live Deployment URL** | https://kotak-alaskxntxna.replit.app (Replit, the required sponsor platform) and https://kotak.tp073654.workers.dev (Cloudflare Workers, the live demo, with Kopi's voice) |
| **Codebase Repository URL** | https://github.com/TolongLabs/Kotak |
| **Name (Leader)** | Hee Zi Jie |
| **Email & phone (Leader)** | heezijie2004@gmail.com, +60125118986 |

Submitted with this report: the pitch deck, **submission/SDC2026_T010NG_PitchDeck.pdf** (also **.pptx**), and the backup pitch video, https://youtu.be/HQgdbantbN0 (README, Links). Both URLs above returned HTTP 200 at 1:45 AM MYT on 1 November 2026.

## 2. Project Description *[Maximum 250 words]*

*246 of 250 words*

Kotak shows a Malaysian worker which of their skills are rising or declining with AI, and gives them a dated, cited plan for what to learn next. It serves SDG 4, target 4.4: more youth and adults with relevant skills for decent jobs.

In 2024, 39.8% of Malaysians aged 15 and over had above-basic digital skills in all five areas (UN SDG indicator 4.4.1). 56% of Malaysian professionals use AI tools at work, but only 26% have had formal training (Hays Asia Salary Guide 2025). Workers get static skills lists with no dates, no scores and no plan.

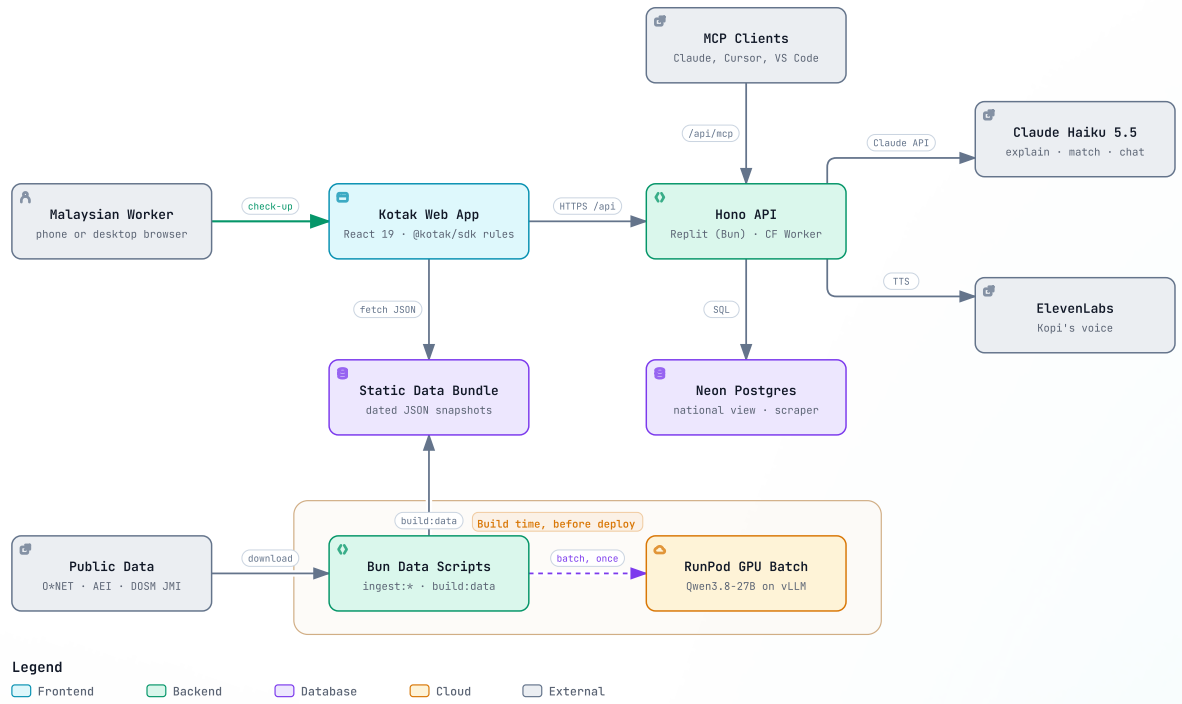
Kopi, a cat counsellor, asks where the user is now, what they studied and what they want, then 12 questions about their job or target role. Kotak draws a Future Skills Radar of the job's 6 core skills, each with a verdict (Rising, Steady, Declining, Split or Unknown) and its dated sources. It ranks up to 6 roles by a fit score that shows its sum. It builds an ordered roadmap with each step's hours, one free resource and its SDG targets. It covers 923 jobs, in English, Malay and Chinese.

Fixed rules over dated public data set every verdict, score and step: the Anthropic Economic Index for Malaysia (April and May 2026), DOSM Job Market Insights and O*NET 30.1. AI only does language work, and a verifier rejects any explanation that changes a verdict. When May's data replaces April's, verdicts change and What Changed lists each one.

# 3. UPGRADED PRODUCT & TECHNICAL EXECUTION

## 3.1. Final System Architecture

*Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.*




*Kotak's architecture, from **docs/readme/architecture.json** (README, Architecture). Build time runs once before deploy; the check-up runs in the browser.*

Kotak is one TypeScript monorepo on Bun. The core rule: **the data decides; the AI explains**. No engine function calls a model, and no model output can change a number (**docs/algorithm-card.md**).

- **Frontend**: React 19, Vite and Tailwind CSS, with Recharts for the radar, three.js for the 3D space of all 923 jobs and Live2D for Kopi. The rule engine, **@kotak/sdk**, runs in the browser over a static bundle of dated JSON snapshots, one file per job, each Zod-checked at build time. A check-up needs no server, so it still works when every API is down.
- **Backend**: a Hono API, the same build on Bun (Replit) and on Cloudflare Workers, with a KV response cache and one Durable Object for chat and MCP. Every route validates its request and response with Zod.

# 3. UPGRADED PRODUCT & TECHNICAL EXECUTION

## 3.1. Final System Architecture (continued)

- **GenAI APIs**: Claude Haiku 5.5 matches a job title typed in any language to an O*NET job, which the user confirms; explains verdicts and the What Changed summary; runs Ask Kopi, an agent with 8 read-only tools that compute every number with the engine; and judges a certificate uploaded as proof of a roadmap step. A verifier (**apps/api/src/ai/verifier.ts**) rejects text that changes a verdict or drops a dated citation, and a fixed template answers instead. Other assistants reach the same tools at **/api/mcp**. ElevenLabs gives Kopi a voice; failed speech falls back to text.
- **Database**: Neon Postgres 18 (**aws-ap-southeast-1**) holds the national view (a role code, 6 skill levels and a date, with real totals only for 10 or more check-ups), the JobStreet scraper's runs, the job ads that admins reviewed, and admin accounts (technical declaration 3.2). Job ads sit beside the verdicts and set none of them.
- **Build-time data**: Bun scripts turn O*NET 30.1, the Economic Index and DOSM Job Market Insights into dated snapshots. Each skill's AI trend resolves Malaysia first, at the finest level published, before any global figure. A RunPod serverless GPU ran Qwen3.8-27B on vLLM once to group each job's own O*NET tasks into 6 skills and draft its can-do statements, for 923 jobs with none failed; a validator rejects any task from another job (README, How We Use RunPod). A weekly GitHub Actions job rebuilds the snapshots and opens a pull request that a person approves.
- **Deployment**: GitHub Actions deploys to Cloudflare Workers on a merge to **main**, and Replit publishes the same build from **main**. Keys live in host secrets, never in the repository.

**SkillsFuture Singapore Skills Framework** (**data/sgf/**, **scripts/sgf/link.ts**). O*NET is a US list, so Kotak adds the region's employer-validated names and levels as context. Claude Haiku 5.5 mapped the 2,030 Singapore job roles to 304 O*NET jobs (510 high, 1,197 medium and 323 low confidence; low matches are not shown). It then linked up to 3 SkillsFuture skills, with proficiency levels 1 to 6, to 1,328 of the 1,612 skill corners in 269 jobs; 125 carry an "emerging" skill. The run took 592 requests, about US$0.56. A validator rejects any code outside O*NET and drops any skill outside the job's own matched roles. Every match is labelled auto-generated, and no rule reads it, so no verdict changes. For the demo job, Web Developers, the Singapore equivalent is Front End Developer, and "Technical documentation" links Test Planning at level 3 of 6. The team uses the framework on verbal approval from a SkillsFuture ambassador on 31 October 2026. SWDA's terms of use ask for prior written permission, which the team does not have yet (**docs/TRD.md**).

# 3. UPGRADED PRODUCT & TECHNICAL EXECUTION

## 3.2. Implementation & Hackathon Deliverables *[Maximum 300 words]*

*Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.*

Numbers such as #460 name pull requests and issues in the team's development repository, TolongLabs/Kotak-dev, which is private. Each commit replayed into the public repository carries a **Dev-Commit:** trailer with the matching development commit's SHA.

*274 of 300 words*

**Built in the sprint, from 9:30 AM on 31 October; each pull request is merged and live**

- **Proof of each roadmap step** (#460 to #462): the user uploads a certificate, and Claude Haiku 5.5 rules pass, fail or unclear. Only a pass marks the step done, and the file is never stored.
- **SkillsFuture Singapore** (#464, #465): each skill's regional name and expected level, beside the verdicts and never read by them.
- **Interpret** (#466): each radar gap card asks Ask Kopi what the gap is, why it matters and how to start. Live replies took 8.6 s and 9.3 s.
- **Verdict tooltips** (#466): a first-time user had no way to learn what Rising, Steady, Declining, Split or Unknown means. Hovering a chip now says it.
- Ask Kopi as a centred modal, a plan bar that fades before Step Details, and a date-picker fix (#466, #467).
- Malay and Chinese content per job (#454), a translated 3D space (#447) and a sortable jobs table (#450).

**Improved from the Phase 1 design, from feedback (Updated PRD)**

- Every SDG chip names its target, gives a reason and the UN text, and the roadmap totals them (#405, #413). We rejudged 1,838 of 5,520 skill tags against the UN text (#414).
- The roadmap scales each step's hours by situation, experience and qualification (#363, #364).
- 902 jobs ask what-would-you-do questions instead of self-ratings (#415).
- A published cost answers "how will Kotak keep running?" (#393, #380).
- A paired-profile test checks that profile fields don't move verdicts or fit (#379).

**Pivot:** the employer team view stays unbuilt. We present it as the next step, not as a feature (Updated PRD, "Not changed, and why").

# 4. ENVIRONMENTAL & SDG IMPACT VERIFICATION

*Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.*

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

| Metric Category | Your Response |
|---|---|
| **Target SDG Metric Achievement**<br>*Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers. e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."* | **Phase 1 KPI**, the sourced-roadmap rate: the share of check-ups that reach a roadmap in under 3 minutes where every verdict, fit score and milestone shows its source and date, with 0 uncited recommendations (Preliminary, section 4).<br>• **Demo-role target**: the April-to-May 2026 swap changes at least one verdict on 100% of the hand-checked demo roles. **Achieved: 3 of 3**, each with 6 of 6 verdicts changed and its roadmap reordered.<br>• **Baseline against now**: on April 2026 data alone, 0 of the 5,520 skills of 923 jobs had a verdict. On May 2026 data, 5,519 of 5,520 have one, and all 923 jobs changed at least one verdict.<br>• **Sourced and fast**: each demo check-up lists its dated sources (16, 14 and 15) and a free https link for each of its 23 roadmap steps. The engine ran all three check-ups, with What Changed, in 0.44 s.<br>• **Fairness**: 30 of 30 paired profiles that differ only in profile fields get identical verdicts and fit.<br>• **20 seeded profiles**, a team measurement that the submitted repository cannot verify: 20 of 20 reached a sourced roadmap, with 0 uncited. Its counter script lands only after the hackathon.<br>Indicator 4.4.1 (39.8% in 2024) cannot move within a hackathon, and we claim no change to it. A pilot (section 5) would measure steps done and fit gained. |
| **Verification Evidence Method**<br>*How was this data verified? e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.* | • **Rerun on the submitted repository**, after **bun install**: **bun run build:data** builds the bundle and checks each demo role's April-to-May change; **bun scripts/sdk-checkup.ts** runs a whole check-up per demo role through the SDK, checks it against the SDK's Zod schemas and prints its sources and What Changed; **bun scripts/scan-verdict-flips.ts** scans all 923 jobs; **bun test packages/sdk/src/fairness.test.ts** checks the 30 pairs. We ran all four on 1 November 2026. Output in 4.1.<br>• **By hand**: **docs/algorithm-card.md** states every formula and **docs/data-card.md** every source and date, so any number can be recomputed.<br>• **Cost**: token counts measured on the running API, and one browser walk, on 29 October 2026 (**docs/cost-card.md**). |

# 4. ENVIRONMENTAL & SDG IMPACT VERIFICATION

| Metric Category | Your Response |
|---|---|
| **Socio-Economic / Civic Value**<br>*How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?* | • **Free, with no account**: the check-up runs in the browser, so one more user adds no server maths.<br>• **Checkable by a partner**: every number cites a dated public source, so a career centre, TVET provider or HR team can check a recommendation by hand before advising on it.<br>• **People decide**: no role is hidden for a missing qualification; a profession Malaysian law regulates shows its route in and its regulator beside the fit.<br>• **Fair by design**: the rules have no field for name, gender, race, religion or age.<br>• **Open to partners' systems**: Ask Kopi's 8 tools serve any assistant at **/api/mcp**, and the national view gives agencies totals only for groups of 10 or more.<br>• **Three languages**: English, Bahasa Malaysia and Chinese behind one switch. |
| **Deployment & Commercial Feasibility**<br>*e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.* | • **Deployed** on Cloudflare Workers and Replit from one build; a local build runs offline.<br>• **As a plugin**: the MCP endpoint **/api/mcp** and the **@kotak/sdk** engine let a career portal or HR system run the same check-up inside its own product, and **plugins/kotak/** ships the check-up as a Claude Code skill.<br>• **Cost per check-up**: US$0.072 with Kopi's voice, US$0.021 muted. A month costs US$71.80 for 1,000 users and US$6,972.51 for 100,000 (US$1,876.51 muted). The voice is about 70% of it (**docs/cost-card.md**).<br>• **Revenue, not yet tested**: free for workers; employers pay per seat for a team view that is not built yet; agencies pay yearly for anonymous totals (**docs/demo/final/qa-cheat-sheet.md**). |

**Personal data under the PDPA 2010** (Technical & Tools Declaration, 6.1). Kotak follows the seven principles of Malaysia's Personal Data Protection Act 2010 (Act 709, sections 5 to 12) and holds no data that names a user (General, Data Integrity). There are no user accounts; the sign-in page says guest mode keeps the check-up on this device. The job, goals, profile and answers stay in the browser's **localStorage**, and Settings > Reset All deletes them (Notice and Choice, Retention, Access). The app sends a typed job title and a role's skill data to the Claude API, Ask Kopi's messages and check-up tool results to Claude, and Kopi's lines to ElevenLabs; the app adds no name, email or ID, and chats are never cached or logged (Disclosure). An uploaded proof file goes to Claude once to be judged and is never stored or logged; only the verdict stays in the browser. Keys stay in host secrets (Security). Postgres holds the national view as a role code, skill levels and a date, with real totals only for 10 or more check-ups; no screen sends them yet.

# 4.1. MATHEMATICAL PROOF / CODE LOGIC SNIPPET

## 4.1. Mathematical Proof / Code Logic Snippet *(Optional, but holds bonus marks)*

*Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.*

*This proof is graded across three categories at once: **Impact and Feasibility** (does the math support your claim), **Technical Execution** (does it reflect what you actually built), as well as **Innovation and Creativity** (is the approach original).] Do not leave this section generic or copied from Phase 1.*

**A. The rules that set every number (packages/shared/src/rules/, excerpts)**

```
// scoring.ts: level 0 to 3 from the basic and advanced answers (1 to 4)
export function cornerLevel(basic: Rating, advanced: Rating): number {
  return (1.5 * (basic - 1)) / 3 + (1.5 * (advanced - 1)) / 3
}
/** fit_score = sum(min(user_level, target_level)) / sum(target_level) * 100, with the maths exposed. */
export function fitScore(corners: { corner_id: string; user_level: number; target_level: number }[]): FitScore
{
  const terms = corners.map((c) => ({ ...c, term: Math.min(c.user_level, c.target_level) }))
  const numerator = terms.reduce((sum, t) => sum + t.term, 0)
  const denominator = terms.reduce((sum, t) => sum + t.target_level, 0)
  return { score: (numerator / denominator) * 100, numerator, denominator, terms, is_estimate: true }
}
// verdicts.ts: the 5-row verdict table. Null (shown as Unknown) when an input is missing.
export function verdictRow(trend: AiTrend, mode: CollaborationMode, demand: DemandState): VerdictRow | null {
  if (trend === 'unknown') return null
  if (trend === 'rising') {
    if (mode === 'unknown') return null
    return mode === 'augmentation' ? 'Rising: AI helps it scale' : 'Declining because of AI'
  }
  if (demand === 'up') return "Rising: AI can't do it yet"
  if (demand === 'flat') return 'Steady'
  if (demand === 'down') return 'Declining for other reasons'
  return null
}
// trend is 'rising' when the projected 12-month change is above RISING_THRESHOLD = 0.05; the projection is a
// least-squares line through the dated points, and is null (Unknown) with fewer than 2 dated points.
/** target = clamp(2.0 + ai_effect + demand_effect, 1.0, 3.0). Shrinking or unknown AI use adds no AI effect.
*/
const growth = Math.max(0, projection?.change ?? 0)
const size = growth / (growth + AI_EFFECT_HALF_GROWTH)          // AI_EFFECT_HALF_GROWTH = 0.5
const ai_effect = mode === 'unknown' || size === 0 ? 0 : mode === 'augmentation' ? size : -size
```

In formulas: level = (basic + advanced − 2) ÷ 2; fit = Σ min(level, target) ÷ Σ target × 100; target = clamp(2 + ai_effect + demand_effect, 1, 3), with demand_effect +0.5 up, 0 flat, −0.5 down. The fast future moves Malaysia halfway to Singapore's AI use index: (0.87 + ½ × (5.81 − 0.87)) ÷ 0.87 = 3.84 times the growth on May 2026 data (**docs/algorithm-card.md**). This is why April data gives Unknown: one dated point fits no line, so **verdictRow** gets '**unknown**' and returns null.

# 4.1. MATHEMATICAL PROOF / CODE LOGIC SNIPPET

## 4.1. Mathematical Proof / Code Logic Snippet (continued)

**B. Rerunnable on the submitted repository (run 1 November 2026 on dev commit 70381d1; output verbatim)**

**bun run build:data**: the demo roles' check (exit 0)

```
Demo 15-1254.00 Web Developers: 6 verdict flips, roadmap reordered, ranking same, Malaysian detailed corners
6/6, May Declining, Steady, Rising, Steady, Rising, Steady
Demo 15-1255.00 Web and Digital Interface Designers: 6 verdict flips, roadmap reordered, ranking moved,
Malaysian detailed corners 6/6, May Steady, Steady, Rising, Steady, Rising, Declining
Demo 43-3031.00 Bookkeeping, Accounting, and Auditing Clerks: 6 verdict flips, roadmap reordered, ranking same,
Malaysian detailed corners 6/6, May Declining, Declining, Declining, Rising, Declining, Declining
```

**bun scripts/sdk-checkup.ts**: a whole check-up through **@kotak/sdk** for each demo role, answers "basic 3, advanced 1" on every skill (exit 0; 0.44 s of wall time for all three roles under **time**). The Web Developers block, with the roadmap after step 3 and the source list cut at "…":

```
Web Developers (15-1254.00), snapshot 2026-05, slow scenario
Asked first: Technical documentation: "I can record simple test results with guidance."
  Web site building: level 1 (self-reported), target 1.2, Declining because of AI. AI use for this work is
rising and AI mostly does it alone. Job-ad demand is flat.
  Web testing and updates: level 1 (self-reported), target 2.0, Steady. AI use for this work is not rising and
AI mostly does it alone. Job-ad demand is flat.
  Technical documentation: level 1 (self-reported), target 2.9, Rising: AI helps it scale. AI use for this work
is rising and AI mostly helps people do it. Job-ad demand is flat.
  Web design and databases: level 1 (self-reported), target 2.0, Steady. AI use for this work is not rising and
AI mostly helps people do it. Job-ad demand is flat.
  Web specifications and standards: level 1 (self-reported), target 2.8, Rising: AI helps it scale. AI use for
this work is rising and AI mostly helps people do it. Job-ad demand is flat.
  Web performance and support: level 1 (self-reported), target 2.0, Steady. AI use for this work is not rising
and AI mostly helps people do it. Job-ad demand is flat.
Best fits of 21 roles:
  1. Document Management Specialists 51% (estimated), declining; gaps: Compliance Monitoring, Policy Standards
  2. Computer Network Architects 50% (estimated), steady; gaps: Network design and specs, Network maintenance
and upgrades
  3. Blockchain Engineers 50% (estimated), steady; gaps: Blockchain architecture, Data integration
Roadmap for Web Developers: 670 h at 4 h a week (user), finish 2030-01-10; the AI dividend frees 5.9 h a week
  Weeks 1-13 (2026-10-25 to 2027-01-23): Level 2 in Technical documentation: "I can write detailed test plans
and procedures independently." https://developers.google.com/tech-writing
  Weeks 13-28 (2027-01-17 to 2027-05-08): Level 2 in Web testing and updates: "I can evaluate code for browser
and device compatibility." https://developer.mozilla.org/en-US/docs/Learn_web_development/Extensions/Testing
  Weeks 28-45 (2027-05-02 to 2027-09-04): Level 2 in Web specifications and standards: "I can write detailed
descriptions of web site specifications." https://developer.mozilla.org/en-
US/docs/Learn_web_development/Getting_started/Web_standards
  …
Sources: (16 dated sources, cut here)
What changed 2026-04 to 2026-05: 6 verdict flips, roadmap reordered, ranking reordered
  Web site building: Unknown to Declining because of AI (verdict, target, ai_use, roadmap)
  Web testing and updates: Unknown to Steady (verdict, ai_use)
  Technical documentation: Unknown to Rising: AI helps it scale (verdict, target, ai_use)
  Web design and databases: Unknown to Steady (verdict, ai_use, roadmap)
  Web specifications and standards: Unknown to Rising: AI helps it scale (verdict, target, ai_use)
  Web performance and support: Unknown to Steady (verdict, ai_use, roadmap)
```

# 4.1. MATHEMATICAL PROOF / CODE LOGIC SNIPPET

## 4.1. Mathematical Proof / Code Logic Snippet (continued)

**bun scripts/scan-verdict-flips.ts**: every job, April to May (exit 0; the AI-use resolution counts and the top-15 list are cut at "...")

```
923 occupations scanned with batch corners (923 in O*NET).
Occupations with at least one April → May verdict flip: 923
Flips by type: {
  "Unknown → Rising": 2064,
  "Unknown → Steady": 2063,
  "Unknown → Declining": 1383,
  "Unknown → Split": 9,
}
May verdicts: {
  Rising: 2064,
  Steady: 2063,
  Declining: 1383,
  Split: 9,
  Unknown: 1,
}
…
Roadmap reordered: 861; fit-score ranking reordered: 684
```

**bun test packages/sdk/src/fairness.test.ts**: 33 pass, 0 fail

```
Fairness: 30 paired profiles on 3 demo roles: verdicts and each role's fit identical in 30 of 30; ranking
differs by design (seniority and qualification rules) in 6; pace differs by design in 30; name, gender, race,
religion and age are rejected as inputs.
```

### C. Team measurement, not verifiable in the submitted repository

A KPI counter script reviewed in the team's development repository, and not merged before the deadline, ran the 20 seeded profiles on 1 November 2026. It counts a verdict as sourced only when a dated source of its own stands behind it, besides the two Economic Index rows every job shares. Its summary lines, on May and then April 2026 data:

```
20/20 profiles reached a sourced roadmap. Engine time: slowest 107 ms, median 49 ms, every one inside the 3-
minute budget.
Recommendations: 120 verdicts, 636 ranked-role fit scores and 161 milestones, each with a dated reading of its
own. Uncited: 0. Counted apart, as Unknown: 0.

0/20 profiles reached a sourced roadmap. Engine time: slowest 202 ms, median 63 ms, every one inside the 3-
minute budget.
Recommendations: 0 verdicts, 0 ranked-role fit scores and 0 milestones, each with a dated reading of its own.
Uncited: 0. Counted apart, as Unknown: 882.
```

### What the proof shows

- **Impact and feasibility**: the verdicts follow the data. With one dated month, every skill reads Unknown; with May's second month, 5,519 of 5,520 skills get a verdict from dated rows that each check-up lists.
- **Technical execution**: every line above is the shipped engine on the shipped data, rerun with one command each, and the fairness test shows profile fields leave verdicts and fit unchanged.
- **Innovation**: a 5-row table over dated rows sets each verdict, not a model, so anyone can recompute any number by hand from the cited source.

# 5. MARKET VIABILITY & FUTURE SCALABILITY PLAN

*SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?* ***[Maximum 250 words]***

*225 of 250 words*

Kotak already scales on cost. The rules run in each user's browser over one shared monthly snapshot, so more users add no server maths. A check-up costs about US$0.072, and 100,000 users cost about US$6,973 a month (**docs/cost-card.md**). With funding or partners, we would scale in four steps.

1. **Pilot.** Through the Rotary network, run Kotak with HR teams at levy-paying employers, and with university career centres and TVET providers. Measure steps done and fit gained per learner: the evidence indicator 4.4.1 needs.
2. **Data at volume.** Get SWDA's written permission for the SkillsFuture data. Load TalentCorp MyCOL and live job-ad counts. Have people review the AI-drafted skills, starting with the most-checked jobs. Translate all 923 jobs. Attach checked, HRD Corp-claimable courses to each step.
3. **Infrastructure.** Raise the 30-calls-a-minute Claude limit and load-test it. Give each user their own Durable Object. Pre-generate each job's question audio, which cuts the voice cost from US$0.051 to US$0.018 a check-up. Move to paid Workers and Neon Launch.
4. **Revenue.** Workers stay free. Employers pay per seat for Kotak Team, a skills-gap view of a whole team, which is not built yet. Agencies pay yearly for anonymous regional totals. These prices are a test, not a contract.

Joint research at XMUM could test the verdict rules against hiring: do skills Kotak calls Rising appear in more Malaysian job ads a year later?

# 6. FINAL SUBMISSION CHECKLIST

*Tick all boxes that apply before submitting.*

- [x] The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
  Both links returned HTTP 200 at 1:45 AM MYT on 1 November 2026. Every key is optional, so the app also builds and runs locally (README, Getting Started).
- [x] The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
  README.md: Getting Started, Prerequisites, Installation (Replit or local) and Run Tests.
- [x] The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
  TolongLabs/Kotak held 246 commits on 1 November, 75 of them since 28 October; the team's work is replayed into it commit by commit with the original authors and dates (**scripts/submission/port-commits.ts**).
- [x] The final pitch deck has been exported and saved as a flattened PDF file.
  **submission/SDC2026_T010NG_PitchDeck.pdf**, with the **.pptx** beside it.
- [x] All fields in this template have been completed and word limits respected.
  Section 2: 246 of 250 words; 3.2: 274 of 300; 5: 225 of 250. Also in **submission/**: the Preliminary proposal, the Updated PRD, the Technical & Tools Declaration and the Replit and RunPod proofs of use.

## 7. IP & Originality Declaration

**7.1. Pre-existing / prior-owned work.** Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No ☑ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it: not applicable. The Phase 1 answer was No: Kotak's code, design and research are new work by the team during the hackathon.

# 7. IP & ORIGINALITY DECLARATION

**7.2. Third-party code, datasets, or models.** Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed). New since Phase 1: SkillsFuture, the ESCO and CIP crosswalks, the JobStreet ads, Claude Sonnet 5.5 and three.js.

| Name | Type (code/ dataset/model) | License |
|---|---|---|
| O*NET 30.1 Database | dataset | CC BY 4.0, with O*NET's credit line |
| Anthropic Economic Index, April and May 2026 | dataset | Data CC BY; code MIT |
| DOSM Job Market Insights; OpenDOSM underemployment | dataset | DOSM terms, with source credit; CC BY 4.0 |
| UN SDG indicator 4.4.1, Malaysia | dataset | UN terms of use; one baseline cited |
| CIP 2020 to SOC 2018 Crosswalk; ESCO to O*NET-SOC Crosswalk | dataset | Public domain (US government); CC BY 4.0 |
| SkillsFuture Singapore Skills Framework 2.0 | dataset | SWDA terms of use, which ask for prior written permission; verbal approval from a SkillsFuture ambassador on 31 October 2026, nothing in writing yet |
| JobStreet job listings, collected by the team's scraper | dataset | JobStreet's terms apply; Kotak publishes totals only |

# 7. IP & ORIGINALITY DECLARATION

**7.2. Third-party code, datasets, or models** (continued)

| Name | Type (code/ dataset/model) | License |
|---|---|---|
| Claude Haiku 5.5, through the Claude API | model | Anthropic Commercial Terms of Service |
| ElevenLabs **eleven_v4** and **eleven_v4_turbo** | model | ElevenLabs Terms of Service |
| Qwen3.8-27B-FP8, run on RunPod with vLLM | model | Apache 2.0 |
| Claude Sonnet 5.5, through the Claude API (rechecked the SDG tags) | model | Anthropic Commercial Terms of Service |
| Live2D Hijiki sample model and Cubism Core | model (2D art) | Live2D Free Material License Agreement; Cubism Core under Live2D's own licence |
| Bun, TypeScript, React, Vite, Tailwind CSS, Recharts, three.js, Hono, Zod, the Anthropic SDK, the MCP SDK, the Neon driver, Wrangler, Playwright, Clearcote, Lucide | code | Open-source licences (MIT, Apache 2.0, ISC and others), each listed in **NOTICE.md** |

Build-time drafts (the what-would-you-do questions, explainers and reversed wording) were also written with Claude Sonnet, GPT, Gemini and DeepSeek models, checked by scripts and labelled auto-generated. The team wrote code with AI coding tools: Claude Code, and Codex CLI, OpenCode and Antigravity CLI with GPT, DeepSeek, GLM and Gemini models; a team member reviewed every change. The full list, with versions, is in the Technical & Tools Declaration, sections 2 to 5.

**7.3. Employment-related IP conflicts.** Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No ☑ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns: not applicable. The Phase 1 answer was No.

# 8. DECLARATION

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.



Signature: Hee Zi Jie, team leader, T010NG

1 November 2026

Date